The University of the South Pacific

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School of Information Technology, Engineering, Mathematics and Physics

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CS211: Computer Organisation

Test 2 – Semester 1, 2024

Time Allowed: 50 mins

Total Marks: 30 (15% of final grade)

Name: _____ Student ID: _____

INSTRUCTIONS:

- There are 8 questions in this paper and all are compulsory.
- All answers must be written in the spaces provided in this booklet.
- You are allowed to use a silent, non-programmable calculator.
- Show all working.

1. A CPU has a data bus width of 16 bits and an address bus width of 20 bits. What is the maximum amount of memory that can be addressed by the CPU? (4 marks)

   Answer:

   The maximum amount of memory that can be addressed by a CPU is determined by the size of its address bus. In this case, the CPU has an address bus width of 20 bits, which means it can address up to 2^20 or 1,048,576 memory locations. Each memory location is 16 bits wide, so the maximum amount of memory that can be addressed is 1,048,576 x 16 bits or 16,777,216 bits, which is equivalent to 2 MB of memory.
2. A CPU is designed to have the following instructions:

   - 3 two-address instructions
   - 30 one-address instructions
   - 32 zero-address instructions

   The machine code length for each instruction is 12 bits. Each address field is 5bits. Using the expanding opcode technique, list the machine codes for the above instructions and determine whether the design is possible or not. (5 marks)

   ANS:
   00 A1 A2
   01 A1 A2
   10 A1 A2

   11 00000 A2
   .
   .
   .
   11 11110 A2

   11 11111 00000

   11 11111 11111 => possible
3. Assume you have a computer which currently does not have any cache memory. Main memory access time is 150 nsec and disk access time is 2500 nsec. The memory hit ratio is 60%. A faster memory is always accessed before a slower one. You want to add a cache that has access time of 90 nsec such that the average access time for your computer becomes 365 nsec. What should you set as the cache hit ratio to achieve this average access time? (5 marks)

   ANS:

   Let p = cache hit ratio

Avg access time:

p x 90 + 0.6 x (150+90) + (0.4 – p) x (2500 + 150 + 90) = 365

90 p + 144 + 1096 – 2740 p = 365

-2650 p = -875

p = 0.3302 = 33.02%

4. For the disjunctive normal form given below:

$$f(w,x,y,z) = \bar{w}\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + \bar{w}x\bar{y}z + \bar{w}xyz + \bar{w}xy\bar{z} + w\bar{x}\bar{y}\bar{z} + w\bar{x}y\bar{z}$$

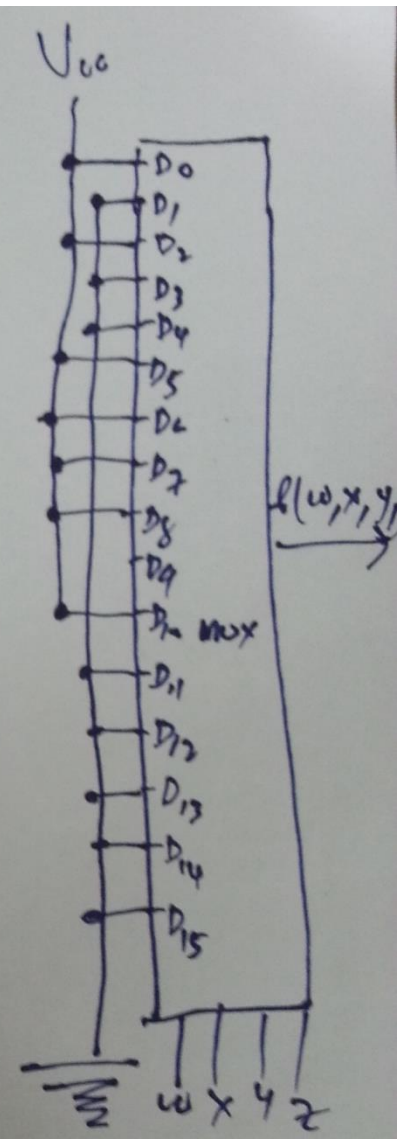
   a) Draw the truth table for *f(w,x,y,z)*. (2 marks)
   b) Use the truth table to draw the multiplexor chip for *f(w,x,y,z)*. (3 marks)
   c) Simply the above DNF expression using Boolean Algebra. (3 marks)
   d) Draw the circuit diagram for the simplest form in part c) above. (2 marks)

ANS:

a)

w	x	y	z	f(x,y,z,w)
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

b)



c) $f(x,y,z) =$

$$\underbrace{\bar{w}\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z}} + \bar{w}x\bar{y}z + \bar{w}xyz + \bar{w}xy\bar{z} + w\bar{x}\bar{y}\bar{z} + w\bar{x}y\bar{z}$$

$$\bar{w}\bar{x}\bar{z}\cancel{(\bar{y}+y)} + \bar{w}xz\cancel{(\bar{y}+y)} + \bar{w}xy\bar{z} + w\bar{x}\bar{z}(\bar{y}$$

$$\Rightarrow \bar{w}\bar{x}\bar{z} + \bar{w}xz + \bar{w}xy\bar{z} + w\bar{x}\bar{z}$$

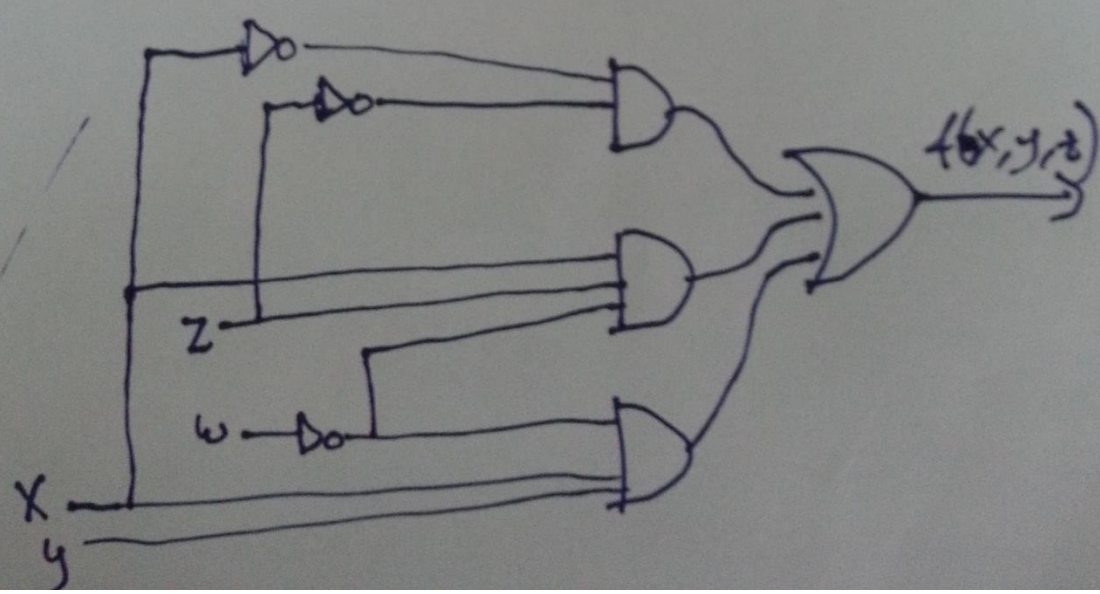
$$\cancel{(\bar{w}+w)}\bar{x}\bar{z} + \bar{w}xz + \bar{w}xy\bar{z}$$

$$\Rightarrow \bar{x}\bar{z} + \underbrace{\bar{w}xz}_{A} + \underbrace{\bar{w}xy\bar{z}}_{B}$$

$$\bar{x}\bar{z} + (\bar{w}xz + \bar{w}x\bar{z})(\bar{w}xz + y)$$

$$\bar{x}\bar{z} + (\bar{w}x\cancel{(z+\bar{z})})(\bar{w}xz + y)$$

$$\bar{x}\bar{z} + \bar{w}xz + \bar{w}xy$$



5. Consider the following infix assignment statement:

```
X = ((A-B)+((C-D)/E))*F
```

Use this statement to answer the following five questions.

a. Using two-operand instructions and no more than 4 registers (R0-R3), show the instructions required to implement the statement. Assume that MOVE A,B moves A to B, ADD A,B adds A to B, SUB A,B subtracts A from B, MUL A,B multiplies A times B, and DIV A,B divides A into B. In all cases, A or B can be a register or memory address, and the result is always stored in B. (2 marks)

b. Assuming each instruction takes 16 bits if only registers are referenced, and 16 additional bits for each memory reference, how many bytes long is your program? (1 mark)

c. Using 0-operand instructions on a stack-oriented machine, show the instructions required to implement the statement. Assume that PUSH A pushes the contents of location A onto the stack, ADD replaces A B with A + B, SUB replaces A B with A - B, MUL replaces A B with A * B, DIV replaces A B with A / B, SWAP replaces A B with B A, and POP A takes the top item off the stack and stores it to location A. (In the above, if A B are on the stack, consider B to be the top of the stack and A to be the one below.) (2 marks)

d. Assuming each instruction takes 8 bits for the opcode and 16 bits for a memory address, if any, how many bytes long is your program? (1 mark)

Answer:

a)

```
MOVE A,R0
SUB B,R0
MOVE C,R1
SUB D,R1
DIV E,R1
ADD R0R1
MUL F,R1
MOVE R1,X
```

b) 7 operands * 2 bytes per operand + 8 instructions * 2 bytes per instruction = 30 bytes.

c)

```
PUSH A
PUSH B
SUB
PUSH C
PUSH D
SUB
PUSH E
DIV
ADD
PUSH F
```

```
MUL
POP X
```

a) 7 operands * 2 bytes per operand + 12 instructions * 1 byte per instruction = 26 bytes.